

Reciprocating Compressor Manual — Model C-200 Series

1. General Description

The C-200 series reciprocating compressors are used for process gas compression in Process Line 2, downstream of the P-102 cooling water pump. Compressor C-201 supplies compressed gas to the downstream separator unit.

2. Technical Specifications

Discharge pressure: up to 12 bar(g). Rated capacity: 1200 Nm³/h. Cooling: water-jacketed cylinders, fed from the same cooling water loop as pump P-102. Maximum discharge temperature: 150 degrees Celsius.

3. Interdependency with Cooling Water Loop

Because C-201 shares its cooling water supply with pump P-102, a drop in cooling water pressure or flow in the shared loop can cause simultaneous temperature increases on both the compressor cylinders and the pump bearings. When both units show elevated temperature at the same time, the root cause is very likely in the shared cooling water system (strainer blockage, cooling water pump issue, or valve misalignment) rather than in either machine individually.

4. Troubleshooting High Discharge Temperature

- Check cooling water flow and pressure to the cylinder jackets.
- Verify suction valve and discharge valve condition (leaking valves increase compression work).
- Check for fouling on cylinder cooling water side (scale buildup reduces heat transfer).
- Confirm compressor is not operating above rated capacity.